

HAOTIAN ZHENG

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Software engineer with expertise in SwiftUI and real-time graphics. Currently at Apple, building UI frameworks used across macOS and iOS. Previously created top-ranked indie apps. Passionate about human-computer interaction and creative tools.

EDUCATION

- **Carnegie Mellon University** Pittsburgh, PA
Master of Science in Mobile and IoT Engineering (GPA 3.91/4.0) January 2021 - May 2022
- **Central South University** Changsha, China
Bachelor of Engineering in Computer Science and Technology Sept 2014 - June 2019

SKILLS

- **Programming Language**
Objective-C, Swift, C#, Java, JavaScript. (2000+ stars on [GitHub](#))
- **Platforms & Skillsets**
Application Dev (iOS/macOS/Android), Game Dev (Unity3D), Web Dev (Vue.js), UI Design (Sketch)

PROFESSIONAL EXPERIENCE

- **Apple** Cupertino, CA
SwiftUI Framework Engineer July 2022 - Present
 - **Development:** Led the design and rollout of several public SwiftUI APIs, working cross-functionally with Human Interface, technical writers, and adopter teams to ensure clarity, consistency, and platform alignment.
 - **Ownership:** Served as the component owner of SwiftUI List, Table, and ScrollView on Mac. Responsible for issue triage, bug fixes, perf optimizations (up to 6x faster in cold launch time), test reliability, and roadmap planning for future improvements.
 - **Presentation:** Presented as the session speaker of [Tailor macOS windows with SwiftUI](#) in WWDC 2024.
- **Apple** Cupertino, CA
Bluetooth Software Intern May 2021 - August 2021
 - **Development:** Built internal Bluetooth audio prototype using CoreAudio, CoreBluetooth, SwiftUI, and XPC, featuring low-latency optimizations and integration with the macOS system UI.
 - **Presentation:** Among top 10% of SWE interns to present project demos and Keynotes to Craig Federighi, SVP of SWE.
- **ret studio** Beijing, China
Co-founder & Chief Engineer September 2018 - January 2019
 - **Development:** Led development of interactive VR pipeline, a REPL-like tool turning movie scripts into 3D visualizations.
 - **Collaboration:** Coordinated closely with other founders for the startup to be admitted to Y Combinator W19 batch.
- **Baidu** Beijing, China
Software Engineer Intern March - August in both 2016 & 2017
 - **Development:** Delivered 3D graphical frontends for smart speakers. Developed native JNI plugins with shared bitmap access for GearVR devices, enabling Android WebView to function as a touchable 3D surface in virtual reality.

PROJECTS

- **Apple WWDC Scholarships** June 2018 - June 2022
 - Won the 2022 challenge with Pegboard, a game logic node editor that that works like UnrealEngine Blueprint but implemented purely in Swift. Designed the system to execute data flow and logic flow with extensible data and UI model. (SwiftUI)
 - Won the 2020 challenge with Shader Node, an interactive GLSL shader playground packed with a node-based interface written from ground-up. Implemented graph structure and searching algorithms to store and traverse nodes. Introduced code generation for GLSL operators and types so shader previews can be updated in realtime. (SpriteKit, UIKit Dynamics, Combine, SwiftUI).
 - Won the 2018 ticket with Golf GO, a golf game written in 1000 lines but provides millions of procedurally generated terrain maps. Achieved runtime mesh modification by applying noise values as vertex offsets onto a plane. (SceneKit, GameplayKit, ModelIO).
- **Unity Native Integration** June 2017 - January 2021
 - Established native plugins on Asset Store to make Unity engine dynamically output to multiple surfaces on Android, mostly utilized in live wallpapers and used by several vendors on 100k+ users.
 - Deployed ContentProvider-based early-init techniques on Android and method swizzling on iOS to safely redirect and dispatch conditional lifecycle calls, greatly extending engine's behavior for non-game use-cases.
- **Indie App Business** October 2016 - January 2021
 - Recreated Apple Music's fluid-like album effect in Android by implementing Kawase Blur and Domain Wrapping in LibGDX. Improved shader performance with lookup table by 80% as measured by RenderDoc / GAPID so low-end devices can run it.
 - Developed, launched, and promoted Android apps once ranked as the most paid app on Google Play (US region, Jan 21, 2018) and covered by The Verge, LifeHacker, and The Next Web.
 - Published iOS apps that were featured on App Store (Apps We Love Right Now) and upvoted as top 4 on Product Hunt.

ACTIVITIES & AWARDS

- **Hackathons:** Multiple hackathon winner record, including 1st place in IKEA hackathon, 1st place in SegmentFault Hackathon, 2nd place in Microsoft & HNU Hackathon, 2nd place in China Academy of Art Hackathon, 3rd place in Uber Hackathon, etc.